

AUTAN-ECG: An AUToencoder bAsed system for anomaly detection in ECG signals

Ugo Lomoio ¹, Patrizia Vizza ², Raffaele Giancotti ², Giuseppe Tradigo ², Salvatore Petrolo ², Sergio Flesca ², Pietro Hiram Guzzi ², and Pierangelo Veltri ²

¹University Magna Graecia of Catanzaro

²Affiliation not available

December 7, 2023

Abstract

Electrocardiographic (ECG) signals that monitor heart activity can help identifying disease-related anomalies. Reliable automatic anomaly detection has been shown to be useful in supporting physicians in reading ECG signals. Decision support systems may be useful in such cases but their reliability can be guaranteed.

Autoencoders (AEs) have been extensively used to analyse signals in many fields. Convolutional Autoencoders (CAE) are a particular class of AE showing optimal performances in detecting signal anomalies. Thus, CAEs can be used to support and automatise the task of anomaly detection. We design and use a CAE-based system to detect anomalies in ECG signals to support cardiologists in identifying anomalies related to possible diseases. Our tool outperforms other state-of-the-art ECG anomaly detection approaches tested on a real dataset. In the task of anomaly detection, our CAE obtains a ROC AUC of 97.82% with a simulated test set and a ROC AUC of 99.75% using on a real test set. The tool and the source code are available at <https://github.com/UgoLomoio/EG.DSS.CAE>.

AUTAN-ECG: An AUToencoder bAsed system for anomaly detectionN in ECG signals

Ugo Lomoio, *Student Member, IEEE*, Patrizia Vizza, Raffaele Giancotti, Giuseppe Tradigo, Salvatore Petrolo, Sergio Flesca, *Member, IEEE*, Pietro Hiram Guzzi, *Member, IEEE*, Pierangelo Veltri, *Member, IEEE*

Abstract—Electrocardiographic (ECG) signals that monitor heart activity can help identifying disease-related anomalies. Reliable automatic anomaly detection has been shown to be useful in supporting physicians in reading ECG signals, for instance in particular conditions such: (i) large data sets such as 24 hour holter recordings or (ii) in the absence of a specialist to read the ECG signals. Decision support systems may be useful in such cases but their reliability can be guaranteed. Autoencoders (AEs) have been extensively used to analyse signals in many fields. Convolutional Autoencoders (CAE) are a particular class of AE showing optimal performances in detecting signal anomalies. Thus, CAEs can be used to support and automatise the task of anomaly detection. We design and use a CAE-based system to detect anomalies in ECG signals to support cardiologists in identifying anomalies related to possible diseases. Our tool outperforms other state-of-the-art ECG anomaly detection approaches tested on a real dataset. In the task of anomaly detection, our CAE obtains a ROC AUC of 97.82% with a simulated test set and a ROC AUC of 99.75% using on a real test set. The tool and the source code are available at https://github.com/UgoLomoio/ECG_DSS_CAE. The CAE based model is able to reconstruct 12-lead ECG windows, simulated and real. It represents a valuable device for clinical as well as for automatic anomaly detection.

keywords: ECG, anomaly detection, signal annotation, autoencoder, decision support systems

I. INTRODUCTION

Biomedical signal analysis supports clinicians in assessing the health and functionality of various organs. ECG signals are used to measure heart activity and identify heart related functional anomalies that can be associated to pathologies or even to predict the risk of heart attack [1]. ECGs are a set of signals from an electrocardiogram which measures the heart's rhythmic electric activity. In the standard ECG, twelve signals are acquired through as many channels to measure electrical activities. The duration of an ECG signal ranges from 10 seconds for standard clinical evaluations up

to 72 hours for cardiac holter continuous monitoring [2]. Moreover, some of the ECG devices are able to measure three additional signals, generating the vectorcardiograms (VCG) that can trace the magnitude and direction of the electrical forces generated by the heart. Wearable devices allow doctors to measure and monitor heart related signals even during activities. Nevertheless, reliable mechanisms able to support clinicians in identify anomalies are still required.

A typical class of anomalies in heart signals are cardiac arrhythmias, i.e. deviations from regular heart rhythms characterised by irregular heartbeat. Electrocardiogram studies allow the detection of fibrillation and frequency anomalies [3], [4]. Clinicians usually identify arrhythmias with a visual examination of the ECG. Such analysis requires a high level of expertise and can be error-prone [5], [6]. Moreover, covert anomalies are by no means easy to discover.

Automatic anomaly identification in ECG signals is a key research area. Computational power of artificial intelligence Decision Support Systems (DSSs) can be used to support clinicians in integrating information obtained from studies and analyses [7]–[11]. DSSs should be able to identify anomalous windows in ECG signals, also through artificial intelligence based models [12]–[14].

Autoencoders (AEs) [15] indicate neural network architectures that have recently gained ground in the research communities focusing on anomaly detection. Transitioning from a standard AE to a Convolutional Autoencoder (CAE) for anomaly detection improves accuracy by enabling the model to capture spatial patterns and relationships in the ECG data, making it more effective at recognising anomalies.

We present a DSS to identify anomalous ECG patterns (i.e. windows) that can be highlighted for signal annotation and diagnosis. The proposed DSS is based on a Convolutional Auto Encoder (CAE) that performs anomaly detection in order to find abnormal windows inside an ECG signal. We tested the DSS on real and synthetic datasets showing that it outperforms other state of the art algorithms for anomaly detection in ECG signals with an accuracy of 99.97% and with a 0.01% Missed Allarm Rate (MAR), where the MAR metric quantifies the error during the classification of anomalous windows in the test set. The obtained MAR value of 0.01% shows that the proposed model can identify almost the totality of anomalous windows present in the test set.

Submitted on November 18, 2023. This work was supported in part by the MISE PON-VQA project.

Ugo Lomoio, Patrizia Vizza, Raffaele Giancotti and Pietro Hiram Guzzi are with Department of Surgical and Medical Sciences, University of Catanzaro, Italy (e-mail: hguzzi@unicz.it)

Giuseppe Tradigo is with SMARTTEST Laboratory of the E-Campus University, Novedrate, Italy

Salvatore Petrolo, Sergio Flesca and Pierangelo Veltri are with the DIMES University of Calabria, Rende, Italy

II. BACKGROUND AND RELATED WORK

The automated analysis of ECG signals is an important research topic since it focuses on extracting information about heart activity or anomalies. We here report algorithmic approaches related to the automatic detection of arrhythmias. Machine Learning applications for anomaly detection have been studied in several research contributions. Li et al. [16] for example, report different machine learning and deep learning models aimed to detect single anomalous heartbeats in ECG signals acquired in health clinics. Such methods were trained and evaluated using the MIT-BIH Arrhythmia Database, which comprises ambulatory 2-channel ECG recordings, and is publicly available since 1989 [17], [18].

While the above-mentioned approaches perform well on the MIT-BIH dataset, they present some drawbacks, such as: (i) the training and testing of these methods rely on outdated 2-leads ECG data (the MIT-BIH), while the current device captures 12 channels and (ii) the need for a segmentation step to extract single heartbeats from the whole signal.

AutoEncoders (AE) perform well in the anomaly detection task. In Liu et al. [19], a Long Short Term Memory AE (LSTM-AE) model is used to perform ECG anomaly detection with an accuracy of 98% with a reconstructed error threshold of 0.02 in terms of absolute error loss. Similarly, available web-based solutions such as Anomagram [20], an interactive website can be used to explore how a Dense AE (DAE) performs on anomaly detection task. The prototype DAE model, tested on the ECG5000 dataset, can predict if a given ECG window is normal or abnormal with an accuracy of 95%. Users can also develop, their own model for anomaly detection on the ECG5000 dataset.

There exist similar application of AE in different contexts. Hou et al. [21] presents a LSTM-AE + Support Vector Machine (SVM) to classify five different types of heartbeats in the MIT-BIH ECG dataset: normal beat, left bundle branch block (LBBB), right bundle branch block (RBBB), atrial premature complexes (APC) and premature ventricular contractions (PVC). The LSTM-AE is used to extract important features from the ECG signals, and then an SVM model uses those features to classify the ECG signals. Their proposed LSTM-AE + SVM model reports the following metrics: an accuracy of 99.74%, a sensitivity of 99.35% and a specificity of 99.84%. In Roy et al. [22] another LSTM-AE model is used to detect abnormalities from ECG signals. The model is tested using the 1-lead ECG5000 dataset, achieving an accuracy of 98.42% with precision, recall and F1 values of 94.52%, 98.22% and 96.33%, respectively. Shan et al. [23] proposed a framework to perform the ECG anomaly detection based on an adversarial AE and temporal convolutional network (TCN) which consist of three modules: an AE, the discriminator, and an outlier detector. The whole framework is evaluated using ECG signals from two different datasets: the MIT-BIH arrhythmia dataset (obtaining an accuracy of 96.76%) and CMUH dataset (with an accuracy 93.58%).

The AE based solution here proposed, is an open source tool with a graphical user interface (GUI) which allows the analysis of long ECG signals (such as the ones obtained

from holter recordings). Differently from existing applications, experimental results reports an efficient time-saving solution for clinicians especially when extensive offline (i.e. not real-time) ECG signal analysis is required. Moreover, comparison of the proposed method with other available tools performing anomaly detection in ECG signals, shows the greater efficiency of the former.

III. MATERIALS AND METHODS

Autoencoders (AEs) [15] refer to neural network architectures that have recently gained wide use in research communities focusing on signal reconstruction, signal denoising and anomaly detection in signals. An AE model includes two different modules as depicted in Figure 1: an Encoder module and a Decoder.

The encoder compresses the acquired data (i.e. ECG windows) and maps them onto a latent space, reducing the data dimensions. The decoder is applied to latent space to map compressed representation back to the original domain (i.e., the input data). The AE model is trained to compress and reconstruct its input using a feedforward approach by minimizing the Mean Square Error (MSE) between the reconstructed output and input. MSE measures the average squared difference between the actual and reconstructed signal as follows:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i represents the actual value, $\hat{y}_i$ the reconstructed value, and n the number of samples in the signal.

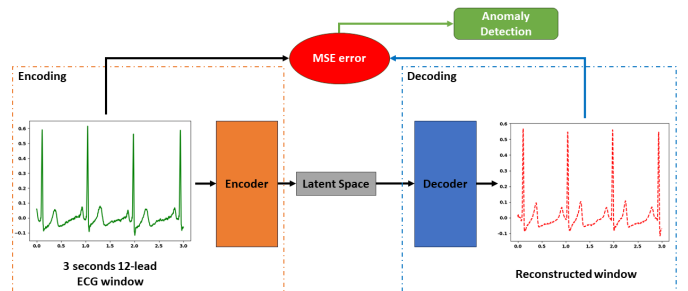


Fig. 1. General AE architecture with: Encoder, Decoder and Latent Space.

To improve our performance in the task at end, we opted for an AE model comprising by convolutional layers, also known as Convolutional AE (CAE). CAE ensures better feature learning, improved signal reconstruction with noise robustness and the preservation of spatial information. An example of a convolutional layer is depicted in Figure 2.

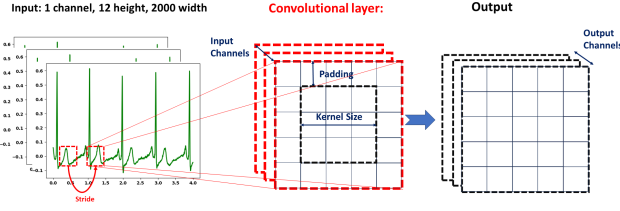


Fig. 2. Input, parameters and output of a convolutional layer.

As the use of convolutional layers ensures reliable features learning, the CAEs perform very well in reconstructing signals or images, presenting noise robustness and the preservation of spatial information. The presented CAE is composed of 2-dimensional convolutional layers for both the encoder and the decoder, while the ending module (the OutputModule) in the decoder is made by 1-dimensional convolutional layers. This architecture allows us to leverage the benefits in processing and reconstructing the input signal. The MSE error between the reconstructed and original signals is then used to perform the anomaly detection [24], [25].

We used both synthetic and real regular 12-lead ECG signal windows to train our CAE, then we tested our model on both separated synthetic and real test sets. Both datasets were first split into normal and abnormal windows. 80% of the normal windows were used to train the model while the remaining 20% was merged with the abnormal dataset to create the test set used for evaluating the model.

Given an ECG signal, the following preprocessing steps are performed: (i) application of a band-filter of 0.5-100 Hz to reject noise such as baseline wander and EMG; (ii) application of a notch filter to attenuate the unwanted 50 Hz interference caused by the electrical grid; (iii) re-sampling of the filtered signal to change the sampling rate from 1000 Hz to 500 Hz; (iv) application of a windowing algorithm to extract 4-second length windows from the ECG signal.

Synthetic ECG signals, used for the testing of the first phase of our CAE, were generated with the SIMARR3 arrhythmia simulator [26]. This software can simulate both normal and atrial fibrillation signals, tachycardia, and bradycardia events at a given sampling rate.

The training simulated set consist of 400 ECG signals with no arrhythmia and without noise (labelled as *normal*) sampled at 500 Hz and with different time duration. The simulated test set consists of: 100 normal ECG signals and 100 ECG signals with atrial fibrillation events (labelled as *anomaly*), sampled at 500 Hz and with different time duration. For both simulated train and test sets, a windowing algorithm has been applied to extract 4-second length signal windows from each signal.

The PTB Diagnostic ECG Database version 1.0 [27] was used for retraining and testing our proposed CAE in a real world scenario. The database contains a total of 549 signals from 290 different patients. Each record includes 15 simultaneously measured channels from two different signals:

- the conventional 12 ECG leads (i, ii, iii, avr, avl, avf, v1, v2, v3, v4, v5, v6);
- the 3 leads VCG (x, y, z).

Each signal in the PTB dataset is digitalized with a sampling rate of 1000 Hz, a resampling step is needed to change the sampling rate from 1000 Hz to 500 Hz. A total of 22 ECG signals of the PTB dataset were excluded because the diagnostic classification of the signal was unavailable. The diagnostic class distribution of the remaining 268 patients is summarised in Table I.

TABLE I
NUMBER OF PATIENTS AND NUMBER OF SIGNALS WITH A GIVEN
DIAGNOSTIC CLASS IN THE PTB DATASET.

Diagnostic class	Number patients	of	Number of signals
Myocardial infarction	148		368
Cardiomyopathy/Heart failure	18		20
Bundle branch block	15		17
Dysrhythmia	14		16
Myocardial hypertrophy	7		7
Valvular heart disease	6		6
Myocarditis	4		4
Palpitation	1		1
Stable angina	1		1
Unstable angina	2		2
Healthy controls	52		80

The extracted ground truth targets were binarized in normal/anomalous, to obtain: 52 normal patients (80 normal signals) and 216 anomalous patients (442 anomalous signals).

Our CAE was trained exclusively on standard 12-lead ECG signal windows, so as to efficiently reconstruct standard ECG windows (with low MSE values) throughout the training process. During the test phase, the CAE encountered challenges rebuilding ECG windows with abnormalities resulting in medium-high MSE values. Difficulties in the reconstruction of anomalous windows were expected since the CAE had not been exposed to signals containing abnormalities during the training phase. Those differences in reconstruction between normal and abnormal ECG windows can be used to classify whether an ECG window is normal or abnormal by means of the AE-based anomaly detection workflow, reported in Figure 3.

The list of MSE errors obtained on the test dataset can be used to select the optimal threshold to adopt in the anomaly detection block depicted in the figure. This parameter is crucial since it estimates CAE performance on the task of anomaly detection which identifies an anomalous window when it observes a reconstruction error above the threshold.

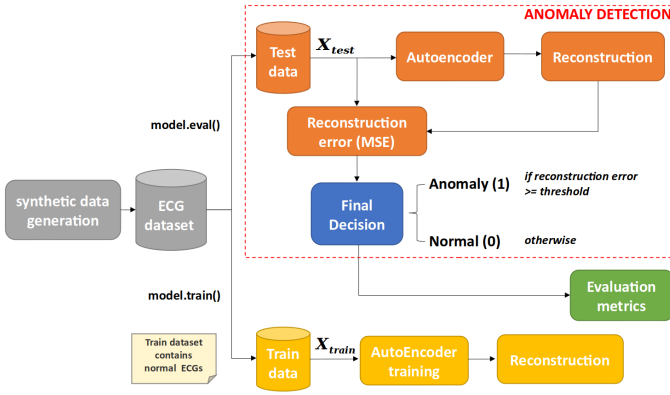


Fig. 3. General workflow for anomaly detection using an AutoEncoder model.

The workflow used to train and test our proposed CAE is depicted in Figure 4.

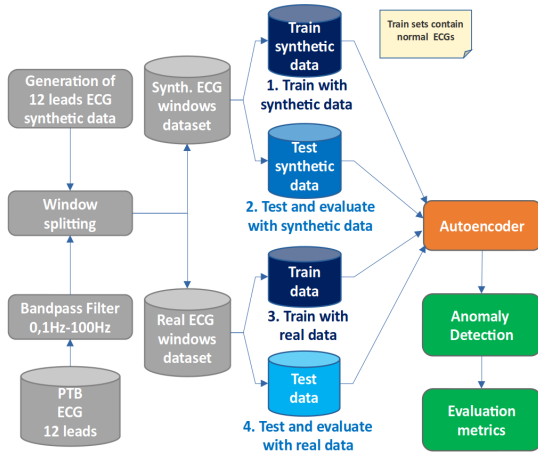


Fig. 4. Workflow used to train and test our proposed CAE. In the first two steps we used only 12-lead synthetically generated ECG windows. Then, we used transfer learning to retrain and evaluate our model in a real world scenario with real ECG signals.

The proposed CAE is depicted in Figure 5. The model input consists of a 4-second signal window of 12 lead ECG sampled at 500 Hz.

The Encoder is composed of the following convolutional blocks:

- *Encoder Block* - Convolutional 2-dimensional layers, Tanh activation function, Batch Normalization layer, Dropout and skip connection layers. Architecture and parameter information is reported in Table II;
- *Last Convolution Encoder Block* - Convolutional 2-dimensional layers and Tanh activation function used as the last block of the Encoder. The architecture and parameters are summarized in Table III;

The Decoder is composed of the following convolutional blocks:

- *Initial Convolution Decoder Block* - transposed Convolutional 2-dimensional layers and Tanh activation function used as the first block of the Decoder. Its architecture and parameters are summarized in Table IV;

- *Last Convolution Decoder Block* - Convolutional 1-dimensional layers and Tanh activation function used as the last block of the Decoder. Its architecture and parameters are summarized in Table V;
- *Decoder Block* - transposed Convolutional 2-dimensional layers, Tanh activation function, Batch Normalization layer, Dropout and skip connection layers. Its architecture and parameters are summarized in Table VI.

TABLE II

THE 2-DIMENSIONAL BLOCK USED TO CONSTRUCT THE ENCODER MODULE. ABBREVIATIONS: BN2D FOR BATCHNORM2D, PAD FOR PADDING, DO FOR DROPOUT

layer	input channel	output channel	kernel	stride	pad	do
Conv2D + Tanh + BN2d	[15,30,30]	[15,30,30]	(3,3)	(1,1)	(1,1)	0.2
Conv2D	[15,30,30]	[30,30,45]	(3,3)	(2,2)	(1,1)	-
skipcon Conv2D BN2d	[15,30,30]	[30,30,45]	(3,3)	(1,1)	(0,0)	-
BN2d + Tanh	[30,30,45]	[30,30,45]	-	-	-	0.2

TABLE III

FINAL 2-DIMENSIONAL CONVOLUTIONAL BLOCK IN THE ENCODER MODULE.

layer	input channel	output channel	kernel	stride
Conv2d + Tanh	60	70	(1,3)	(2,2)
Conv2d + Tanh	70	80	(1,3)	(2,2)
Conv2d + Tanh	80	90	(1,3)	(2,2)
Conv2d + Tanh	90	100	(1,3)	(2,2)
Conv2d + Tanh	100	128	(1,5)	(2,2)

TABLE IV

DECODER INITIAL CONVOLUTION BLOCK. ABBREVIATIONS: CONV2D FOR CONVOLUTIONALTRANSPOSE2D, PAD FOR PADDING.

layer	input channel	output channel	kernel	stride	out pad
ConvT2d + Tanh	128	100	(1,5)	(2,2)	(0,1)
ConvT2d + Tanh	100	90	(1,3)	(2,2)	(0,1)
ConvT2d + Tanh	90	80	(1,3)	(2,2)	(0,1)
ConvT2d + Tanh	80	70	(1,3)	(2,2)	(0,1)
ConvT2d + Tanh	70	60	(1,5)	(2,2)	(0,0)

TABLE V

LAST 1-DIMENSIONAL CONVOLUTIONAL BLOCK IN THE DECODER MODULE.

layer	input channel	output channel	kernel	stride	pad	groups
Conv1d + Tanh	15	15	(5,)	(1,)	(2,)	15
Conv1d + Tanh	15	15	(3,)	(1,)	(1,)	15
Conv1d + Tanh	15	15	(3,)	(1,)	(1,)	15
Conv1d + Tanh	15	15	(3,)	(1,)	(1,)	15

TABLE VI

THE 2-DIMENSIONAL CONVOLUTIONAL TRANSPOSE BLOCK USED TO CONSTRUCT THE DECODER MODULE. ABBREVIATIONS: CONV2D FOR CONVTRANSPOSE2D, BN2D FOR BATCHNORM2D, PAD. FOR PADDING, DO FOR DROPOUT

layer	input	output	kernel	stride	pad	out pad	do
ConvT2d + Tanh + BN2d	[45,30,30]	[30,30,15]	(3,3)	(1,1)	(1,1)	-	0.2
ConvT2d	[45,30,30]	[30,30,15]	(3,3)	(2,2)	(1,1)	(1,1)	-
skipcon ConvT2d + BN2d	[45,30,30]	[30,30,15]	(3,3)	(1,1)	(0,0)	(1,1)	-
BN2d + Tanh	45	45	-	-	-	-	0.2

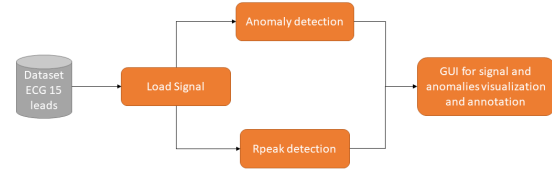


Fig. 7. Workflow of the proposed decision support system tool.

IV. RESULTS

The proposed DSS based on a CAE model has been trained and tested on both synthetic and real ECG data. The CAE was first trained using only normal and free-noise synthetically generated ECG windows signals. It was then tested to discriminate between normal and atrial fibrillation synthetic ECG windows. This was done following the AE-based anomaly detection workflow that uses the MSE reconstruction error and an optimised threshold parameter to make the classification.

The underlying CAE model was first trained on ECG data signals from healthy synthetically generated without noise. We then tested it to verify reliability in terms of anomaly detection to discriminate between normal and atrial fibrillation ECG signals. This was done by optimising the threshold parameter and reconstructing error measurements. In this way the system presents all the windows classified as anomalous.

Given an input 4-second window signal x and its reconstruction $\hat{x}$ in the output of our CAE:

- if the MSE reconstruction error between the first 3 seconds of x and $\hat{x}$ is less than a given threshold parameter, then the window signal is considered normal. Note that the AE takes in input 4 seconds but we used only the first 3 seconds to compute the MSE error and predict the presence of an abnormality. This was due to a poor reconstruction of the AE in the last second present in every ECG window, as can be seen in Figure 8;
- otherwise, the input window signal x is considered anomalous and presented to the clinician for final valuation and annotation.

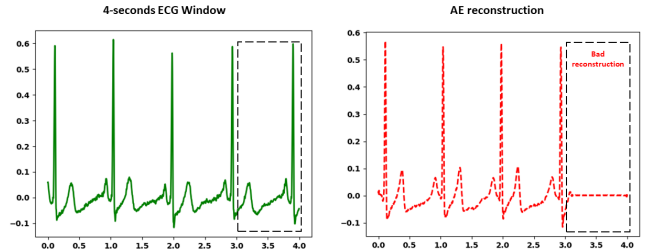


Fig. 8. Example of poor reconstruction in the last second of the signal.

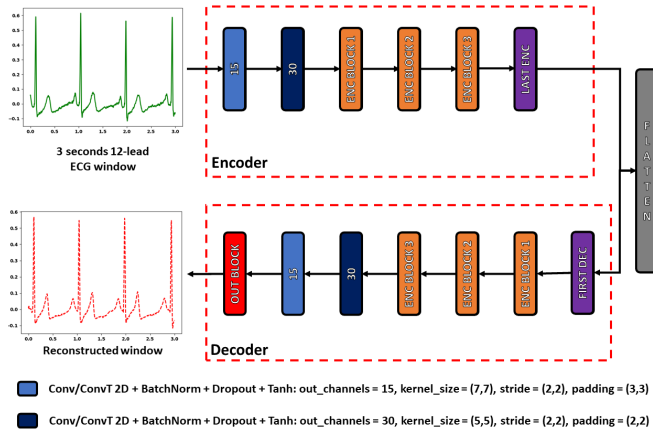


Fig. 5. Proposed CAE architecture.

Finally, we integrate our proposed CAE inside a Python-3 based GUI DSS, depicted in Figure 6.

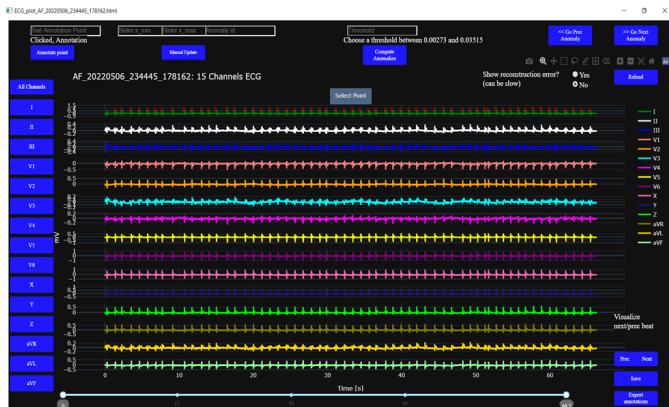


Fig. 6. ECG anomaly detector and initial annotator tool interface.

The workflow of our DDS tool is represented in Figure 7.

Let TP and TN be the number of ECG windows correctly identified as normal or abnormal, respectively, and FP and FN

the number of normal or abnormal ECG windows misclassified, the model has been evaluated in terms of the following performance metrics:

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$
- $Precision = \frac{TP}{TP+FP}$
- $Sensitivity = Recall = \frac{TP}{TP+FN}$
- $F1 = \frac{2*Precision*Recall}{Precision+Recall}$
- $Specificity = \frac{TN}{FP+TN}$
- $False\ Allarm\ Rate = \frac{FP}{TP+FP}$
- $Missed\ Allarm\ Rate = \frac{FN}{TN+FN}$
- ROC AUC is calculated as the Area Under the $Sensitivity-(1-Specificity)$ curve, also known as the ROC curve.

The model was first evaluated and optimised regarding the AUC (Area Under Curve) metric score on the synthetic test set. The AUC is the area under the ROC (Receiver Operating Characteristic) curve. It represents the degree of separability between the two classes, i.e. normal and abnormal, providing an indicator of how well the model can distinguish between two classes. The model scored an AUC equal to 97.82% on the synthetic test set. In Table VII we report a number of classification performance metrics of our CAE in anomaly detection from the synthetic test set. Those metrics are obtained by optimising the threshold parameter to maximise the ROC AUC score. The optimised threshold found is equal to 0.0063.

TABLE VII

PERFORMANCE METRICS OF OUR CAE IN THE ANOMALY DETECTION TASK, USING THE SYNTHETIC DATASET, WITH A THRESHOLD PARAMETER EQUAL TO 0.0063.

Metric name	Metric value
Accuracy	97.19 %
F1 score	94.84 %
Sensitivity	99.14 %
Specificity	96.50 %
ROC AUC	99.72 %
False Allarm Rate	0.86 %
Missed Allarm Rate	3.5 %

After validating our model with a synthetic dataset, we proceeded to validate it on real ECG data. We first tested our synthetic-trained model using all of the real ECG signals contained in the PTB dataset.

A band pass filter 0.5 - 100 Hz was applied to remove noise, then a notch filter was applied to reject the 50 Hz noise from the electrical grid. After, the filtered signal was resampled from a sample rate of 1000 Hz to 500 Hz. Finally, a window splitting algorithm was applied to split the filtered signal in 4-second windows with a 1 second overlap.

The synthetic pretrained model did not perform well enough to be able to analyse real ECG signals with noise, so we retrained the model so as to reconstruct real and noisy healthy ECG signals. We split the PTB dataset in two datasets: one containing only normal and one containing only anomalous signals. We split normal signals in 80% train and 20% test sets. We used the train set to retrain our pre-trained model for 3000 epochs, with a batch size of 4 and an Adam optimizer with 0.0001 learning rate. The final training set contains only healthy real ECG windows (some with noise). The final test set

contains both healthy and unhealthy real ECG windows (some with noise). Train-test split details are reported in Table VIII.

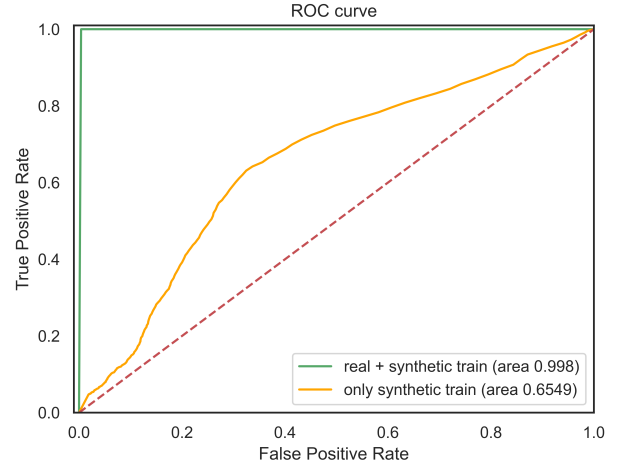


Fig. 9. ROC Curves obtained in the evaluation of our anomaly detection models over the entire PTB dataset.

TABLE VIII
TRAIN TEST SPLIT DETAILS.

Split	Normal windows	Abnormal windows
Train	2398	0
Test	605	15377

Figure 9 shows two ROC curves obtained from two different CAE models trained on the anomaly detection task and tested on the entire PTB dataset:

- the model trained on synthetic and free of noise ECG signals;
- the retrained model that takes into consideration the presence of noise in real ECG signals.

In Table IX we report the real-data classification performance metrics obtained by measuring the retrained CAE on the anomaly detection task. These were obtained by optimising the threshold parameter in order to maximise the ROC AUC score. The optimised threshold found is equal to 0.0026. In Figure 10 are reported the ROC curve and the confusion matrix obtained on the anomaly detection task.

TABLE IX

PERFORMANCE METRICS OF OUR RETRAINED CAE ON THE ANOMALY DETECTION TASK, USING THE TEST SET CONTAINING PTB REAL ECG SIGNALS AND A THRESHOLD PARAMETER EQUAL TO 0.0026.

Metric name	Metric value
Accuracy	99.97 %
F1 score	99.59 %
Sensitivity	99.50 %
Specificity	99.99 %
ROC AUC	99.75 %
False Alarm Rate	0.5 %
Missed Alarm Rate	0.01 %

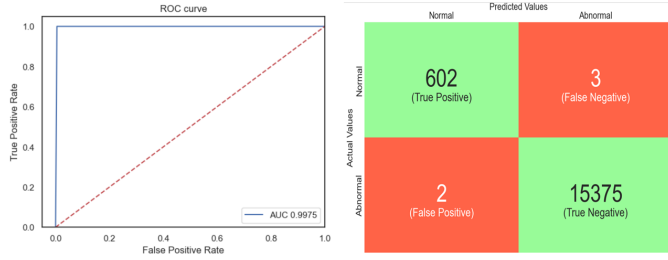


Fig. 10. ROC Curve (left) and Confusion matrix (right) obtained during the evaluation of our model on the anomaly detection task

After this, we evaluate our model on the task of binary classification (normal/abnormal) of each signal singularly. Each signal was preprocessed using the same preprocessing as presented before and then split into ECG 4-second length windows. Each window is classified by means of the previous adopted threshold approach. Finally, if the signal contains at least one window that our CAE classify as anomalous, the entire signal is considered anomalous. In Table X we report some classification metrics of our retrained CAE on the signal classification task using the optimised threshold of 0.0026 used above. Figure 11 reports the ROC curve and the confusion matrix obtained on the signal classification task.

TABLE X

PERFORMANCE METRICS OF OUR RETRAINED CAE ON THE SIGNAL CLASSIFICATION TASK, WITH THE TEST SET CONTAINING PTB REAL ECG SIGNALS AND A THRESHOLD PARAMETER EQUAL TO 0.0026.

Metric name	Metric value
Accuracy	99.56 %
F1 score	93.33 %
Sensitivity	87.50 %
Specificity	100.00 %
ROC AUC	93.75 %
False Alarm Rate	12.50 %
Missed Alarm Rate	0.0 %

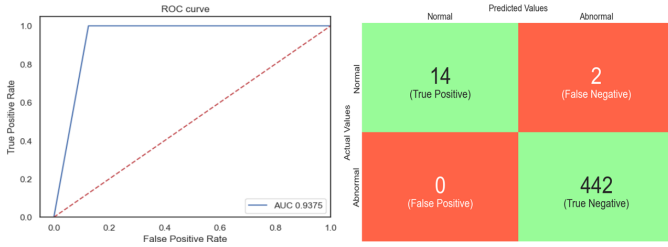


Fig. 11. ROC Curve (left) and Confusion matrix (right) obtained during the evaluation of our model on the signal classification task

In Tables XI and XII we compare our results on the anomaly detection task with various state of the art models carrying out anomaly detection on a real ECG datasets. In Table XIII we report the performance of our proposed CAE in the detection of 12 different types of ECG abnormalities present in the PTB dataset.

TABLE XI

COMPARISON OF THE PROPOSED MODEL WITH RELATED WORKS IN TERMS OF PERFORMANCE (IN BOLD THE HIGHEST SCORES).
Abbreviations: ESTT for European ST-T dataset, MB for MIT-BIH.

Model	Dataset	Nor./Abn.	AUC	Sens.	Spec.	Acc	F1
Ge et al. (2002) [28]	MB	143/713	97.06%	99.02%	93.01%	98.01%	98.81%
Acharya A (2017) [29]	MB	902/20807	47.83%	92.08%	93.13%	92.13%	95.73%
Acharya B (2017) [29]	MB	361/8322	94.58%	99.16%	43.88%	94.90%	97.29%
OC-SVM (2010) [30]	MB	1005/1000	79.40%	80.50%	-	80.40%	78.50%
AnoGAN (2017) [31]	MB	1005/1000	89.40%	91.00%	-	89.10%	92.70%
GANomaly (2018) [32]	MB	1005/1000	93.70%	89.50%	-	90.40%	93.40%
BeatGAN (2019) [33]	MB	1005/1000	94.30%	90.70%	-	92.90%	94.80%
ECG-ADGAN (2023) [34]	MB	1005/1000	95.90%	91.80%	-	94.30%	95.55%
Ullah et al. (2021) [35]	MB+PTB	-	-	99.30%	-	-	99.30%
Iftene et al. (2022) [36]	MB+PTB	-	-	-	-	-	98.00%
Li et al. (2022) [37]	MB+ESTT	2653/2653	99.74%	99.74%	99.85%	99.79%	99.79%
Our proposal (2023)	PTB	605/15377	99.74%	99.50%	99.99%	99.97%	99.59%

TABLE XII

COMPARISON OF PROPOSED MODEL WITH RELATED WORKS IN TERMS OF FALSE ALARM RATE AND MISSED ALARM RATE, DATASET USED AND METHOD.

Model	Dataset	Nor./Abn.	FAR	MAR
Ge et al. [28] (2002)	MIT-BIH	143/713	0.98%	6.99%
Acharya Net A [29] (2017)	MIT-BIH	902/20807	7.91%	6.87%
Acharya Net B [29] (2017)	MIT-BIH	361/8322	0.84%	56.12%
Li et al. [37] (2022)	MIT-BIH + European ST-T	2653/2653	0.26%	0.15%
Our proposal (2023)	PTB	605/15377	0.5%	0.01%

TABLE XIII

CLASSIFICATION METRICS OF THE PROPOSED MODEL IN THE DETECTION OF DIFFERENT TYPE OF ABNORMALITIES, WITH A FIXED AND OPTIMISED RECONSTRUCTION ERROR THRESHOLD OF 0.0026.

Class	Nor./Abn.	AUC	Sens.	Spec.	Acc	F1	FAR	MAR
All	605/15377	99.74%	100%	99.50%	99.97%	99.59%	0.5%	0.01%
Bundle branch block	605/607	99.75%	99.50%	100%	99.75%	99.75%	0.5%	0.0%
Cardiomyopathy	605/587	99.75%	99.50%	100%	99.75%	99.75%	0.5%	0.0%
Dysrhythmia	605/393	99.75%	99.50%	100%	99.70%	99.75%	0.5%	0.0%
Heart failure	605/30	99.75%	99.50%	100%	99.51%	99.75%	0.5%	0.0%
Hypertrophy	605/265	99.56%	99.50%	99.62%	99.54%	99.67%	0.5%	0.038%
Miocardial Infraction	605/13163	99.75%	99.50%	100%	99.98%	99.75%	0.5%	0.0%
Miocarditis	605/152	99.42%	99.50%	99.34%	99.47%	99.67%	0.5%	0.66%
Palpitation	605/10	99.75%	99.50%	100%	99.51%	99.75%	0.5%	0.0%
Stable angina	605/47	99.75%	99.50%	100%	99.54%	99.75%	0.5%	0.0%
Unstable angina	605/10	99.75%	99.50%	100%	99.51%	99.75%	0.5%	0.0%
Valvular heart disease	605/113	99.75%	99.50%	100%	99.58%	99.75%	0.5%	0.0%

Once the CAE was trained and tested using the PTB dataset, we obtained a model to perform ECG anomaly detection on a 12-lead ECG dataset. In order to achieve good performances and a low FAR, we suggest the tool with a 12-lead ECG signal containing few artefacts.

V. DISCUSSION AND LIMITATIONS

The proposed tool uses a deep learning model whose architecture has been optimized to work with 12 ECG leads (i.e. signals). The main limitation of this method is that it restricts the anomaly detection only to ECG signals having 12 leads. In its present form, the tool acts as a signal annotator software for ECG signals having a number of leads other than 12.

However, there are no limitations regarding the duration of the ECG signals due to the time window splitting approach.

Furthermore, one known limitation is the inability to detect cardiac abnormalities in highly noisy ECG signals. Even if the decision support system applies a filtering step to reject noise, it is possible that a noisy but normal ECG window will be detected as abnormal by our CAE. This could be seen as a problem because the tool will register, to the clinician, as abnormal a window that in reality is quite normal but with an abnormality caused by the presence of noise.

Our experiments, using the PTB dataset, report a FAR of 0.5% and an extremely low MAR of 0.01%:

- With a FAR of 0.5% we can say that when classifying normal windows our model makes few errors probably caused by the presence of noise in some ECG signals in the PTB dataset;
- With a MAR of 0.01% we can say that our model makes very few errors when classifying anomalous windows.

So, we can conclude by saying that our tool will present as abnormal:

- quite all real abnormalities in the signal, thanks to a very low MAR;
- few normal windows with noise presence, these are the false abnormalities caused by a low FAR.

We would like to remind the reader that our tool aims only to support the clinician by presenting all the abnormalities found in the signal. That said, is always the clinician that validates the DSS prediction (is it a real abnormality or is it caused by noise?) and who performs the final annotation.

In Figure 12 we report a normal ECG window with noise, misclassified as abnormal by our CAE, which is caused by the presence of a loud noise in the beginning of the window (see the reddish zone on the left part of the Figure).

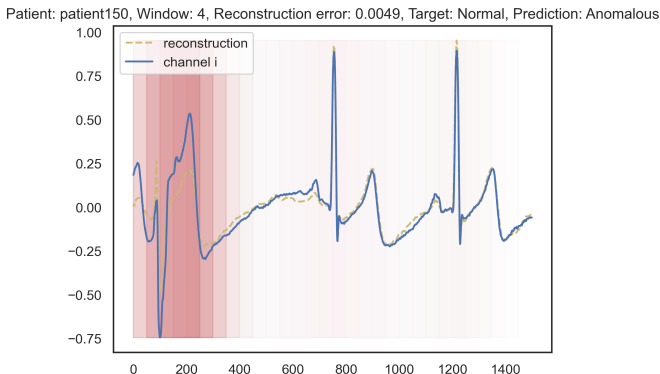


Fig. 12. An example of a misclassification of a normal window caused by the presence of noise in the signal.

VI. CONCLUSIONS

In this work, we have presented a CAE model that can be used to automatically identify anomalies in ECG signals.

As a first step, the AE was trained by means of synthetic and noise-free normal ECG windows. We evaluated the model using synthetic normal and abnormal ECG windows (i.e. containing atrial fibrillation events). For these steps, synthetic signals were generated by means of the SIMARR3 software. We measured the performance of the CAE with synthetic signals on the anomaly detection tasks obtaining an AUC score of 97.82%.

As a second step, we tested the above mentioned model (i.e. trained on synthetic noise-free data) on real ECG signals from the PTB dataset, which gave poor results in terms of accuracy. This was caused by the presence of noise in the PTB dataset. We then retrained our model using 80% of normal signals in the PTB dataset. The remaining 20% of normal signals, together with all of the abnormal signals were used to create the test set. We then measured the performance of the retrained CAE, using the previously created test set, on the anomaly detection tasks obtaining an AUC score of 99.74% and a MAR of 0.01%. Our model equals, and in some classification metrics outperforms, the state-of-the-art model for the anomaly detection in ECG signals.

We then implemented a tool for the loading, analysis, visualization, manual (i.e. guided by the CAE predictions) and semi-automatic annotation of ECG signals.

Our tool can automatically identify anomalous portions of the signal with high precision, and send them to physicians or specialists. Anomalies are correlated with atrial fibrillation events, myocardial infarction, or noise disturbance in the analyzed ECG signal.

In this study, we have shown a tool for the automatic detection of anomalies in ECG signals. However, there remain some limitations to which we plan to address in future works. To overcome the number of channels limitations of our decision support system we intend to train a CAE on ECG signals with 3 leads. We shall also develop a new model able to classify anomalous windows obtained by our CAE (e.g., abnormal windows containing myocardial infarction or a cardiomyopathy events). Thanks to such a model, not only will we be able to present, to the clinician, all abnormal windows inside an ECG signal, but will also be able to suggest a possible diagnosis. These future works will lead to a better understanding of anomaly events in ECG signals and also help optimise clinicians' workflow, potentially leading to the development of new diagnostic or therapeutic tools.

DECLARATIONS

Availability of data and materials

Pretrained models, setup, code, user guide presented or used in this work are available at: https://github.com/UgoLomoio/ECG_DSS_CAE.

Train and test data will be made available upon request.

Declaration of Competing Interest

Authors declare that they have no competing interests.

Funding

PHG, PV, UL and RG were partially funded by the MISE PON-VQA project. UL Ph.D. fellow is partially funded by Relatech S.p.A.

Contributions

UL was responsible of software implementation, testing, and experiments. PV worked at conceptualization, clinical data analysis and experimental discussion. RG participated to testing definition and conceptualization. GT supervised to editing, conceptualization and experimental tests. SP and SF designed the AE structure and applications. PhG and PVe supervised all the processes and edited the manuscripts. All authors read and approved the manuscript.

Acknowledgements

Authors would like to thank Dr. Elena Succurro for her suggestions about the ECG signals. Sergio Flesca, Pietro Hiram Guzzi and Patrizia Vizza have been partially supported by the Next Generation EU - Italian NRRP, Mission 4, Component 2, Investment 1.5, call for the creation and strengthening of 'Innovation Ecosystems', building 'Territorial R&D Leaders' (Directorial Decree n. 2021/3277) - project Tech4You - Technologies for climate change adaptation and quality of life improvement, n. ECS0000009. Pierangelo Veltri was partially supported by project SERICS (PE00000014) under the MUR National Recovery and Resilience Plan funded by the European Union - NextGenerationEU.

- [1] Krishnan Sridhar and Athavale Yashodhan. Trends in biomedical signal feature extraction. *Biomedical Signal Processing and Control*, 43:41–63, 2018.
- [2] Kaplan Berkaya, Selcan Uysal, Alper Kurşat Gunal, Efnan Ergin, Semih Gunal, and Serkan Gulmezoglu. A survey on ecg analysis. *Biomedical Signal Processing and Control*, 43:216–235, 2018.
- [3] Bhatla Anjali, Mayer Michael, Adusumalli Srinath, Hyman Matthew, Oh Eric, Tierney Ann, Moss Juwann, A. Chahal Anwar, Anesi George, Denduluri Srinivas, Domenico Christopher, Arkles Jeffrey, Abella Benjamin, Bullinga John, Callans David, Dixit Sanjay, Epstein Andrew, Frankel David, Garcia Fermin, and Deo Rajat. Covid-19 and cardiac arrhythmias. *Heart Rhythm*, 17, 2020.
- [4] Pietro Hiram Guzzi and Marinka Zitnik. Editorial deep learning and graph embeddings for network biology. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 19(02):653–654, 2022.
- [5] Ke Xu, Shunan Guo, Nan Cao, David Gotz, Aiwen Xu, Huamin Qu, Zhenjie Yao, and Yixin Chen. EcgLens: Interactive visual exploration of large scale ecg data for arrhythmia detection. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, pages 1–12, 2018.
- [6] Greg Wood, Jeremy Batt, Andrew Appelboam, Adrian Harris, and Mark R Wilson. Exploring the impact of expertise, clinical history, and visual search on electrocardiogram interpretation. *Medical Decision Making*, 34(1):75–83, 2014.
- [7] Cairns Andrew, Bond Raymond, Finlay Dewar, Guldenring Daniel, Badilini Fabio, Libretti Guido, Peace Aaron, and Leslie Stephen. A decision support system and rule-based algorithm to augment the human interpretation of the 12-lead electrocardiogram. *Journal of Electrocardiology*, 50, 2017.
- [8] Raza Ali, Tran Kim Phuc, Koehl Ludovic, and Li Shujun. A Survey and Perspective, Decision Support Systems for Healthcare based on Probabilistic Graphical Models, 2021.
- [9] S. Adewole, Mojeed Hamed, Ogunmodede James, Gabralla Lubna, Faruk Nasir, A. Abdulkarim, Ifada Emmanuel, Folawiyos Yusuf, Oloyede Abdulkarim, Abiodun Olawoyin, Sikiru Ismael, Musa Nehemiah, Gital Abdulsalam, and Chiroma Haruna. Expert system and decision support system for electrocardiogram interpretation and diagnosis: Review, challenges and research directions. *Applied Sciences*, 12, 2022.
- [10] Simegn Gizeaddis, Gebeyehu Worku, and Zelalem Mizanu. Computer-aided decision support system for diagnosis of heart diseases. *Research Reports in Clinical Cardiology*, 13:39–54, 2022.
- [11] Pietro Cinaglia, Giuseppe Tradigo, Pietro H Guzzi, and Pierangelo Veltri. Design and implementation of a telecardiology system for mobile devices. *Interdisciplinary Sciences: Computational Life Sciences*, 7:266–274, 2015.
- [12] Liu Xinwen, Wang Huan, Li Zongjin, and Qin Lang. Deep learning in ecg diagnosis: A review. *Knowledge-Based Systems*, 227:10718, 2021.
- [13] Ozdalili Selen Işin Ali. Cardiac arrhythmia detection using deep learning. *Procedia Computer Science*, 120:268–275, 2017.
- [14] Giuseppe Tradigo, Salvatore De Rosa, Patrizia Vizza, Gionata Fragomeni, Pietro Hiram Guzzi, Ciro Indolfi, and Pierangelo Veltri. Calculation of intracoronary pressure-based indexes with jlabchart. *Applied Sciences*, 12(7):3448, 2022.
- [15] D. Bank, N. Koenigstein, and R. Giryas. *Autoencoders*. arXiv, 2020.
- [16] Li Hongzu and Boulanger Pierre. A survey of heart anomaly detection using ambulatory electrocardiogram (ecg). *Sensors*, 20:1461, 2020.
- [17] G. B. Moody and R. G. Mark. The mit-bih arrhythmia database on cd-rom and software for use with it. *Proceedings Computers in Cardiology*, 1990:185–188, 1990.
- [18] Binon Teji, Swarup Roy, Devendra Singh Dhami, Dinabandhu Bhandari, and Pietro Hiram Guzzi. Graph embedding techniques for predicting missing links in biological networks: An empirical evaluation. *IEEE Transactions on Emerging Topics in Computing*, 2023.
- [19] Pengfei Liu, Xiaoming Sun, Yang Han, Zhishuai He, Weifeng Zhang, and Chenxu Wu. Arrhythmia classification of lstm autoencoder based on time series anomaly detection, biomedical signal processing and control, volume 71. Part B, 2022:10322, 2022.
- [20] <https://anomagran.fastforwardlabs.com/>. *Anomagran*.
- [21] Borui Hou, Jianyong Yang, Pu Wang, and Ruqiang Yan. Lstm-based auto-encoder model for ecg arrhythmias classification. *IEEE Transactions on Instrumentation and Measurement*, 69(4):1232–1240, 2020.
- [22] Moumita Roy, Sukanta Majumder, Anindya Halder, and Utpal Biswas. Ecg-net: A deep lstm autoencoder for detecting anomalous ecg. *Engineering Applications of Artificial Intelligence*, 124:106484, 2023.
- [23] Lianfeng Shan, Yu Li, Hua Jiang, Peng Zhou, Jing Niu, Ran Liu, Yuan Yuan Wei, Jiao Peng, Huizhen Yu, Xianzheng Sha, and Shijie Chang. Abnormal ecg detection based on an adversarial autoencoder. *Frontiers in Physiology*, 13, 2022.
- [24] R. Giancotti, U. Lomoio, P. Veltri, P. H. Guzzi, and P. Vizza. A machine-learning based tool for bioimages managing and annotation. *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Las Vegas, NV*, pages 2589–2594, 2022.
- [25] An Jinwon and Sungzoon Cho. Variational autoencoder based anomaly detection using reconstruction probability. *Special Lecture on IE*, 2(1):1–18, 2015.
- [26] Andrius Petrenas Sološenko, Andrius Paliakaite, Birute Sornmo, and Vaidotas Leif Marozas. *Model for Simulating ECG and PPG Signals with Arrhythmia Episodes*. 2021.
- [27] R. Bousseljot, D. Kreiseler, and A. Schnabel. Nutzung der ekg-signalndatenbank cardiodat der ptb "über das internet. *Biomedizinische Technik*, 40, 1995.
- [28] D. Ge, N. Srinivasan, and Krishnan SM. Cardiac arrhythmia classification using autoregressive modeling. *Biomed Eng Online*, 2002:1, Nov 2002.
- [29] U. R. Acharya, H. Fujita, O. S. Lih, Y. Hagiwara, J. H. Tan, and M. Adam. Automated detection of arrhythmias using different intervals of tachycardia ecg segments with convolutional neural network. *Information Sciences*, 405:81–90, 2017.
- [30] Woo Soo-Min, Lee Hye-Jin, Kang Bub-Joo, and Ban Sang-Woo. Ecg signal monitoring using one-class support vector machine. In *10*, pages 132–137, AEE, 2010. Proceedings of the 9th WSEAS International Conference on Applications of Electrical Engineering.
- [31] T. Schlegl, P. Seeböck, S. M. Waldstein, U. Schmidt-Erfurth, and G. Langs. *Unsupervised Anomaly Detection with Generative Adversarial Networks to Guide Marker Discovery. Information Processing in Medical Imaging*, volume 10265. IPMI 2017. Lecture Notes in Computer Science Springer, Cham, , 2017.

- [32] S. Akcay, A. Atapour-Abarghouei, and T. P. Breckon. Ganomaly: Semisupervised anomaly detection via adversarial training. In *Asian conference on computer vision*, , pages 622–637, 2018.
- [33] B. Zhou, S. Liu, B. Hooi, X. Cheng, J. Ye, and Beatgan: Anomalous. rhythm detection using adversarially generated time series. *IJCAI pp*, pages 4433–4439, 2019.
- [34] J. Qin, F. Gao, Z. Wang, et al. A novel temporal generative adversarial network for electrocardiography anomaly detection. *Artificial Intelligence in Medicine*, 136:10248, 2023.
- [35] W. Ullah, I. Siddique, R. M. Zulqarnain, M. M. Alam, I. Ahmad, and Raza UA. Classification of arrhythmia in heartbeat detection using deep learning. *Computational Intelligence and Neuroscience*, 19, 2021.
- [36] A. Iftene, A. Burlacu, and D. Gifu. Atrial fibrillation detection based on deep learning models. *Procedia Computational Science*, 207:3752–3760, 2022.
- [37] H. Li and P. Boulanger. Structural anomalies detection from electrocardiogram (ecg) with spectrogram and handcrafted features. *Sensors*, 22:7, 2022.